

[3]

4. Derive the expression of efficiency of transformer.

Section—C

(Long Answer Type Questions)

Note : Attempt any *one* question. Each question carries 20 marks. $1 \times 20 = 20$

1. Explain 3-phase induction motor principle of operation.
2. Explain the general layout of electrical power system.
3. Explain the construction and working principle of PMCC.
4. Explain and derive maximum power transfer theorem.

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4. Differentiate synchronous and induction motor.
5. What is line and phase current ?
6. What are the applications of DC motors ?
7. Write down the torque equation of motor.
8. Explain parallel R-L-C circuit.
9. Differentiate magnetic and electric circuit.
10. Explain Norton's theorem.

Section—B

(Short Answer Type Questions)

Note : Attempt any *two* questions. Each question carries 12 marks. $2 \times 12 = 24$

1. Derive the expression of emf equation of generator.
2. Explain superposition theorem with an example.
3. (a) Explain any power factor improvement method.
(b) Explain single phase dynamometer wattmeter.

EE-1846

**B. Tech. (First Semester)
EXAMINATION, 2025-26
ELECTRICAL ENGINEERING**

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *seven* questions. Each question carries 8 marks. $7 \times 8 = 56$

1. Explain active and passive element.
2. What are apparent and reactive powers ?
3. Explain the power losses of transformer.

P. T. O.